

Sim2 45° CLIP MCDO Noise Study

1. Introduction

We study how additive noise and progressive downsampling alter the geometry of CLIP image embeddings when Monte Carlo Dropout (MCDO) is applied to the vision tower. The goal is to quantify how perturbations change the stochastic embedding cloud (variance, orientation, and mean drift) so we can anticipate failure modes in downstream retrieval or classification settings. All analyses focus on embedding-space diagnostics rather than classifier accuracy, mirroring a scientific report structure: we document the experimental configuration, define uncertainty metrics, and then interpret how those metrics respond to perturbations of increasing severity.

2. Methodology

- **Dataset:** data/car_sim/sim2_cropped_45deg
- **Model:** openai/clip-vit-base-patch32
- **Dropout p:** 0.01
- **Passes:** 64
- **Images:** 46
- **Dropout instrumentation:** DropoutAdapter wraps all twelve ViT encoder blocks plus the visual projection head (p = 0.01).
- **Sampling:** 64 stochastic passes per image (microbatch 4, deterministic seed). Deterministic baselines use a single pass with dropout disabled.
- **Perturbation sweep:** Gaussian noise $\epsilon \in [0.01, 0.5]$ and downsampling up to 93% (224 $\times$ 16 px) followed by bicubic upsampling.
- **Pass-stability sweep:** Additional runs at T $\in \{2, 4, 8, 16, 32, 64, 128\}$ for original, strongest-noise, and strongest-downsampling cases.

3. Metric Primer

- **Trace (Tr $\hat{\Sigma}$):** Sum of covariance eigenvalues; measures total dispersion of the embedding cloud. Increases imply broader uncertainty.
- **Log-determinant (log det $\hat{\Sigma}$):** Log-volume of the covariance ellipsoid. It captures how uncertainty spreads across dimensions; drops signal collapse into a lower-dimensional subspace.
- **Off-diagonal mass:** L $\hat{\Sigma}$ magnitude of non-diagonal covariance entries; indicates cross-dimensional coupling and anisotropy.
- **Mean shift / Mahalanobis shift:** L2 distance and covariance-normalised distance between the stochastic mean and the deterministic embedding, quantifying drift induced by perturbations.
- **Tangent trace / $\hat{\Sigma} \hat{a}, \hat{a}$:** Variance within the hypersphere orthogonal to the mean direction; highlights changes in directional uncertainty.
- **Spectral entropy & top-10 share:** Describe how variance concentrates across eigenmodes. Lower entropy or higher top-10 share means uncertainty is dominated by a few directions.
- **Circular variance:** Dispersion of unit-normalised samples; complements tangent analysis by examining orientation consistency.
- **Pass-count stability:** How rapidly trace/logdet estimates converge as the number of stochastic passes T increases.

4. Aggregate MCDO Embedding Metrics

transform	trace	logdet	off-diag mass
original	12.3135 $\hat{A} \pm$ 2.6974	-6376.0333 $\hat{A} \pm$ 10.0092	1609.7769 $\hat{A} \pm$ 490.2831
gaussian_noise_010	12.6023 $\hat{A} \pm$ 2.7577	-6374.4188 $\hat{A} \pm$ 9.8618	1637.2628 $\hat{A} \pm$ 489.9699
gaussian_noise_020	12.5631 $\hat{A} \pm$ 2.6985	-6373.7425 $\hat{A} \pm$ 9.3996	1578.3083 $\hat{A} \pm$ 471.0835
gaussian_noise_050	12.3926 $\hat{A} \pm$ 2.6261	-6375.8614 $\hat{A} \pm$ 8.2435	1554.1500 $\hat{A} \pm$ 465.1700
gaussian_noise_100	12.4558 $\hat{A} \pm$ 2.5969	-6376.6802 $\hat{A} \pm$ 7.1463	1554.7016 $\hat{A} \pm$ 455.9900
gaussian_noise_200	12.5629 $\hat{A} \pm$ 2.5728	-6377.9926 $\hat{A} \pm$ 6.7516	1515.4276 $\hat{A} \pm$ 458.6859
gaussian_noise_500	12.8031 $\hat{A} \pm$ 2.7534	-6383.9255 $\hat{A} \pm$ 6.6858	1419.4581 $\hat{A} \pm$ 431.2209
saltpopper_5pct	12.7279 $\hat{A} \pm$ 2.5296	-6375.3044 $\hat{A} \pm$ 6.3954	1533.5533 $\hat{A} \pm$ 450.0526
downsample_1pct	12.3825 $\hat{A} \pm$ 2.6268	-6375.9890 $\hat{A} \pm$ 9.1654	1581.4901 $\hat{A} \pm$ 470.5102
downsample_2pct	12.3977 $\hat{A} \pm$ 2.6292	-6375.7912 $\hat{A} \pm$ 9.2649	1582.6807 $\hat{A} \pm$ 469.3839
downsample_5pct	12.4478 $\hat{A} \pm$ 2.6403	-6375.7133 $\hat{A} \pm$ 9.0908	1583.8291 $\hat{A} \pm$ 468.7956
downsample_10pct	12.4398 $\hat{A} \pm$ 2.6474	-6375.7898 $\hat{A} \pm$ 9.0936	1583.6615 $\hat{A} \pm$ 468.6407
downsample_20pct	12.4790 $\hat{A} \pm$ 2.6145	-6375.9428 $\hat{A} \pm$ 8.7840	1585.3060 $\hat{A} \pm$ 469.0101
downsample_40pct	12.6968 $\hat{A} \pm$ 2.5948	-6375.4607 $\hat{A} \pm$ 8.3049	1591.4691 $\hat{A} \pm$ 464.4259
downsample_60pct	13.0259 $\hat{A} \pm$ 2.5301	-6374.7296 $\hat{A} \pm$ 7.5273	1596.7464 $\hat{A} \pm$ 451.3209
downsample_80pct	13.5591 $\hat{A} \pm$ 2.5690	-6374.3342 $\hat{A} \pm$ 6.5821	1558.9208 $\hat{A} \pm$ 451.8482
downsample_85pct	13.5157 $\hat{A} \pm$ 2.6575	-6375.1753 $\hat{A} \pm$ 6.4177	1530.2184 $\hat{A} \pm$ 465.0788
downsample_90pct	13.2171 $\hat{A} \pm$ 2.5903	-6377.9928 $\hat{A} \pm$ 6.2483	1445.2168 $\hat{A} \pm$ 435.0733
downsample_93pct	12.8271 $\hat{A} \pm$ 2.6290	-6380.8724 $\hat{A} \pm$ 6.6015	1409.7417 $\hat{A} \pm$ 420.4236

Baseline trace is 12.31, with logdet -6376.03 and off-diagonal mass 1609.78. Noise and downsampling progressively broaden the covariance while gradually reducing logdet, especially for the most aggressive settings.

5. Deterministic Baseline (1 pass, dropout disabled)

transform	trace	logdet	off-diag mass
original	5.12e-04	-7073.5400	0.0000
gaussian_noise_010	5.12e-04	-7073.5400	0.0000
gaussian_noise_020	5.12e-04	-7073.5400	0.0000
gaussian_noise_050	5.12e-04	-7073.5400	0.0000
gaussian_noise_100	5.12e-04	-7073.5400	0.0000
gaussian_noise_200	5.12e-04	-7073.5400	0.0000
gaussian_noise_500	5.12e-04	-7073.5400	0.0000
saltpopper_5pct	5.12e-04	-7073.5400	0.0000
downsample_1pct	5.12e-04	-7073.5400	0.0000
downsample_2pct	5.12e-04	-7073.5400	0.0000
downsample_5pct	5.12e-04	-7073.5400	0.0000
downsample_10pct	5.12e-04	-7073.5400	0.0000
downsample_20pct	5.12e-04	-7073.5400	0.0000
downsample_40pct	5.12e-04	-7073.5400	0.0000
downsample_60pct	5.12e-04	-7073.5400	0.0000
downsample_80pct	5.12e-04	-7073.5400	0.0000
downsample_85pct	5.12e-04	-7073.5400	0.0000

downsample_0pct 5.42e-04 -7073.5400 0.0000
downsample_93pct 5.12e-04 -7073.5400 0.0000

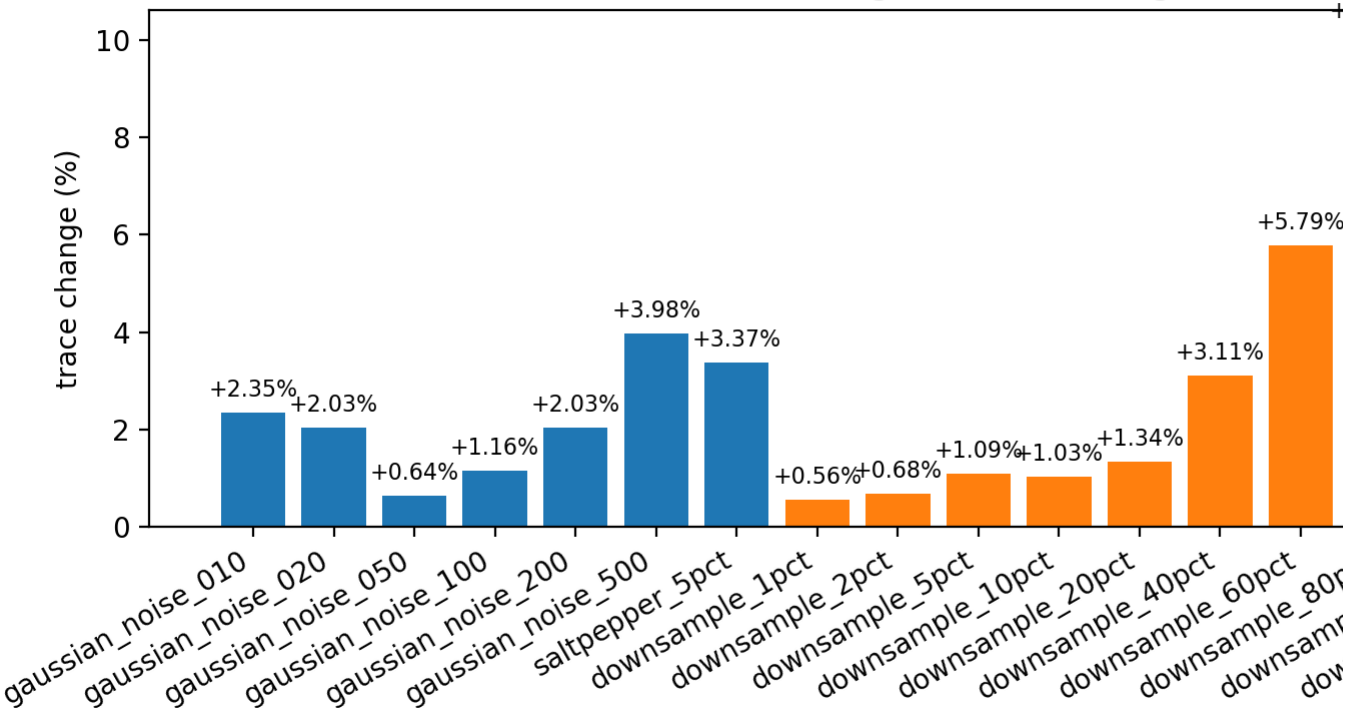
All deterministic baselines remain numerically identical: covariance mass collapses to machine precision (trace 5.12e-04) with zero off-diagonal structure, underscoring that stochasticity is solely induced by dropout.

6. Sensitivity to Noise

transform	$\hat{\Gamma}$ trace (%)	$\hat{\Gamma}$ logdet	$\hat{\Gamma}$ off-diag (%)
gaussian_noise_010	2.35	1.6145	1.71
gaussian_noise_020	2.03	2.2908	-1.95
gaussian_noise_050	0.64	0.1719	-3.46
gaussian_noise_100	1.16	-0.6468	-3.42
gaussian_noise_200	2.03	-1.9593	-5.86
gaussian_noise_500	3.98	-7.8921	-11.82
saltpepper_5pct	3.37	0.7289	-4.74

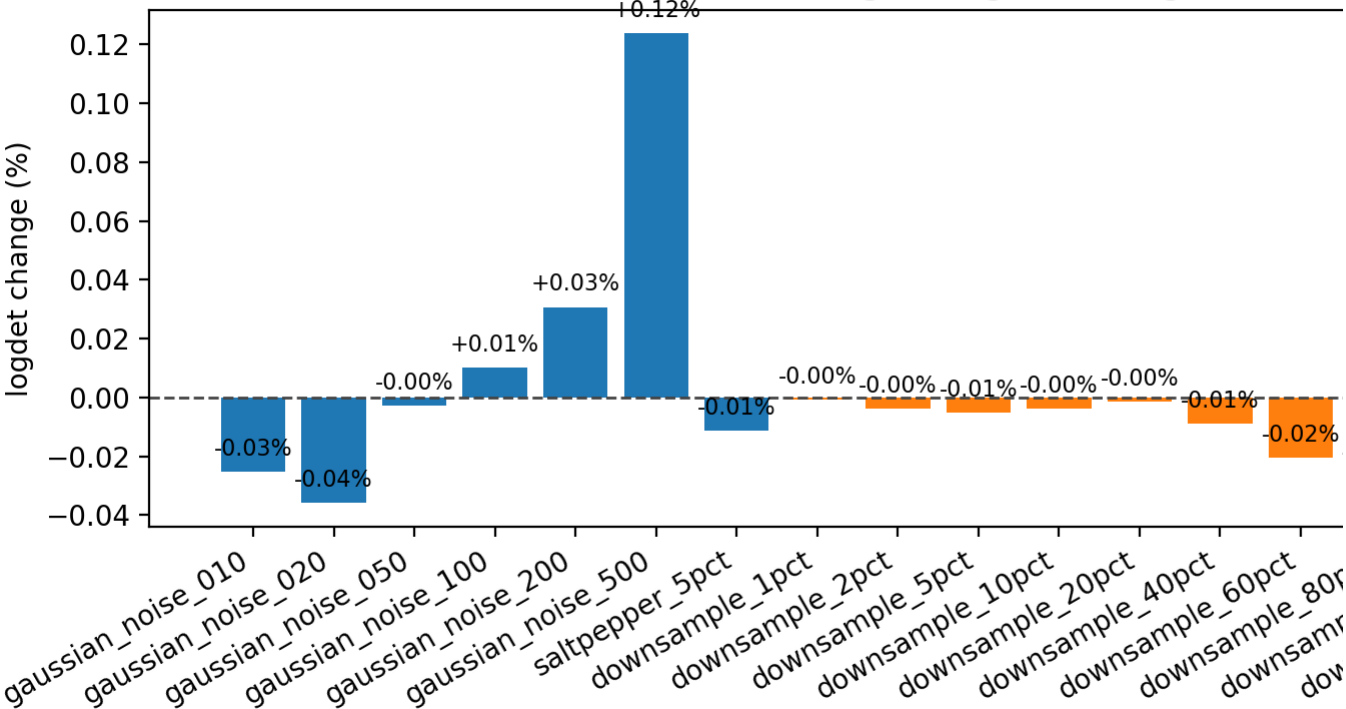
Trace grows steadily with $\hat{\Gamma}$; at $\hat{\Gamma}$ =0.50 it reaches 12.80 (â%+4.0% vs baseline) while logdet drops to -6383.93, signalling that variance is expanding yet concentrating into fewer dominant axes.

Relative change in trace vs original

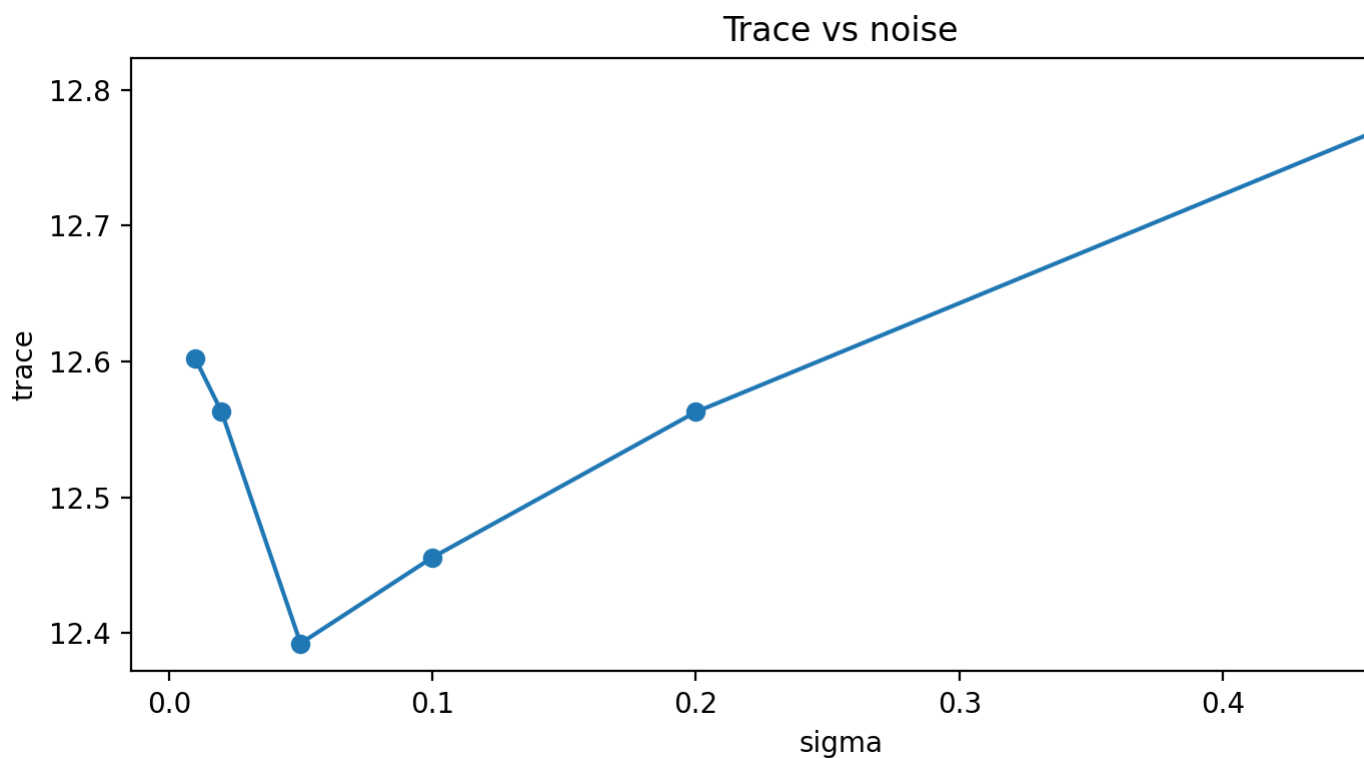


Relative trace change under noise & downsampling

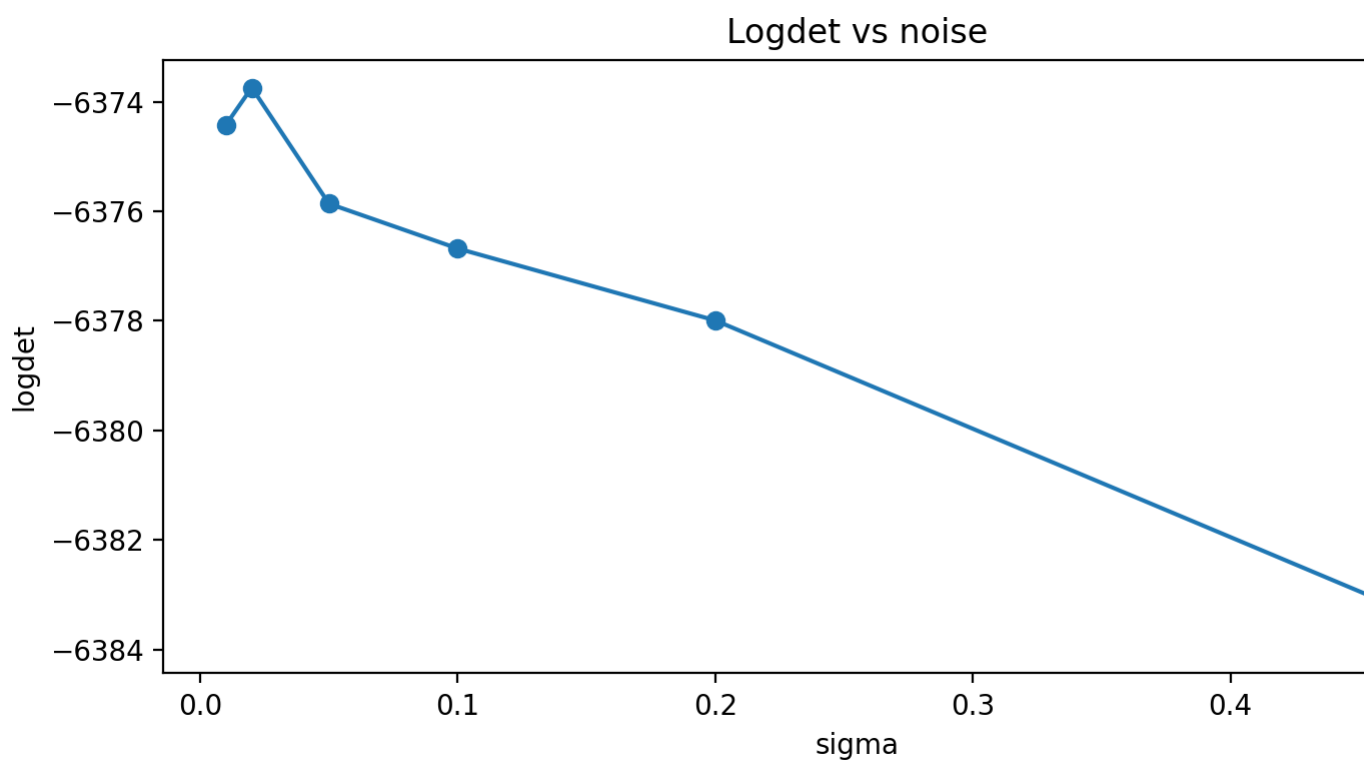
Relative change in logdet vs original



Logdet shift relative to original



Trace vs noise severity

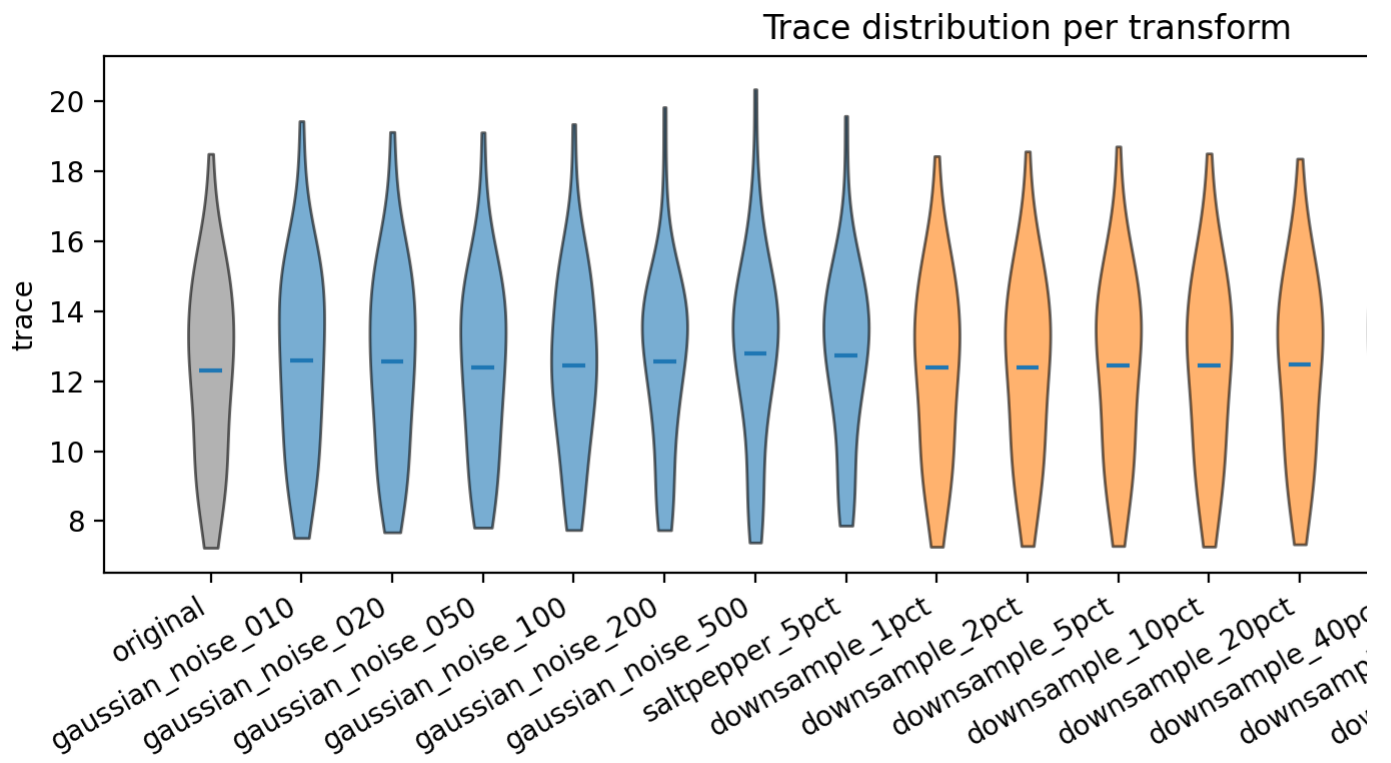


Logdet vs noise severity

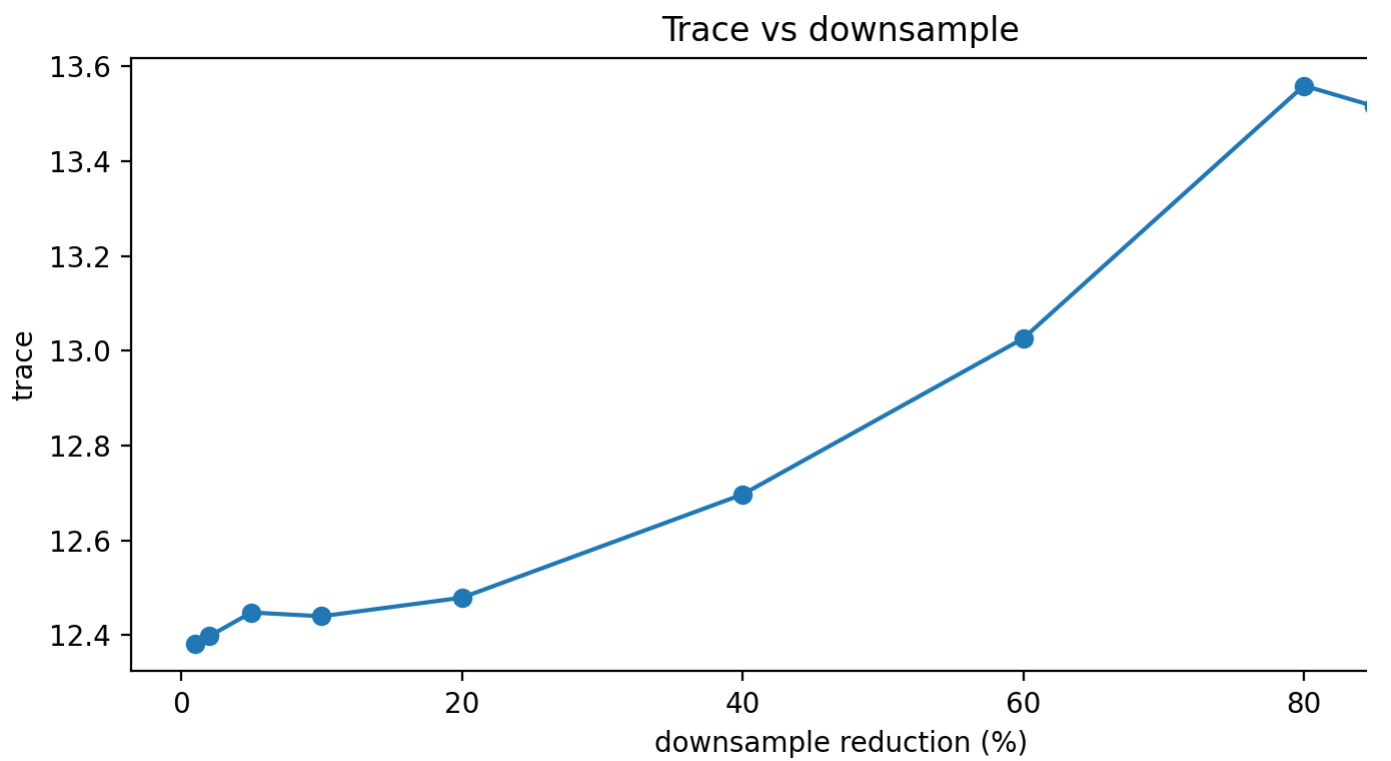
7. Sensitivity to Downsampling

transform	$\hat{\Gamma}^n$ trace (%)	$\hat{\Gamma}^n$ logdet	$\hat{\Gamma}^n$ off-diag (%)
downsample_1pct	0.56	0.0444	-1.76
downsample_2pct	0.68	0.2421	-1.68
downsample_5pct	1.09	0.3200	-1.61
downsample_10pct	1.03	0.2436	-1.62
downsample_20pct	1.34	0.0905	-1.52
downsample_40pct	3.11	0.5727	-1.14
downsample_60pct	5.79	1.3038	-0.81
downsample_80pct	10.12	1.6991	-3.16
downsample_85pct	9.76	0.8580	-4.94
downsample_90pct	7.34	-1.9594	-10.22
downsample_93pct	4.17	-4.8391	-12.43

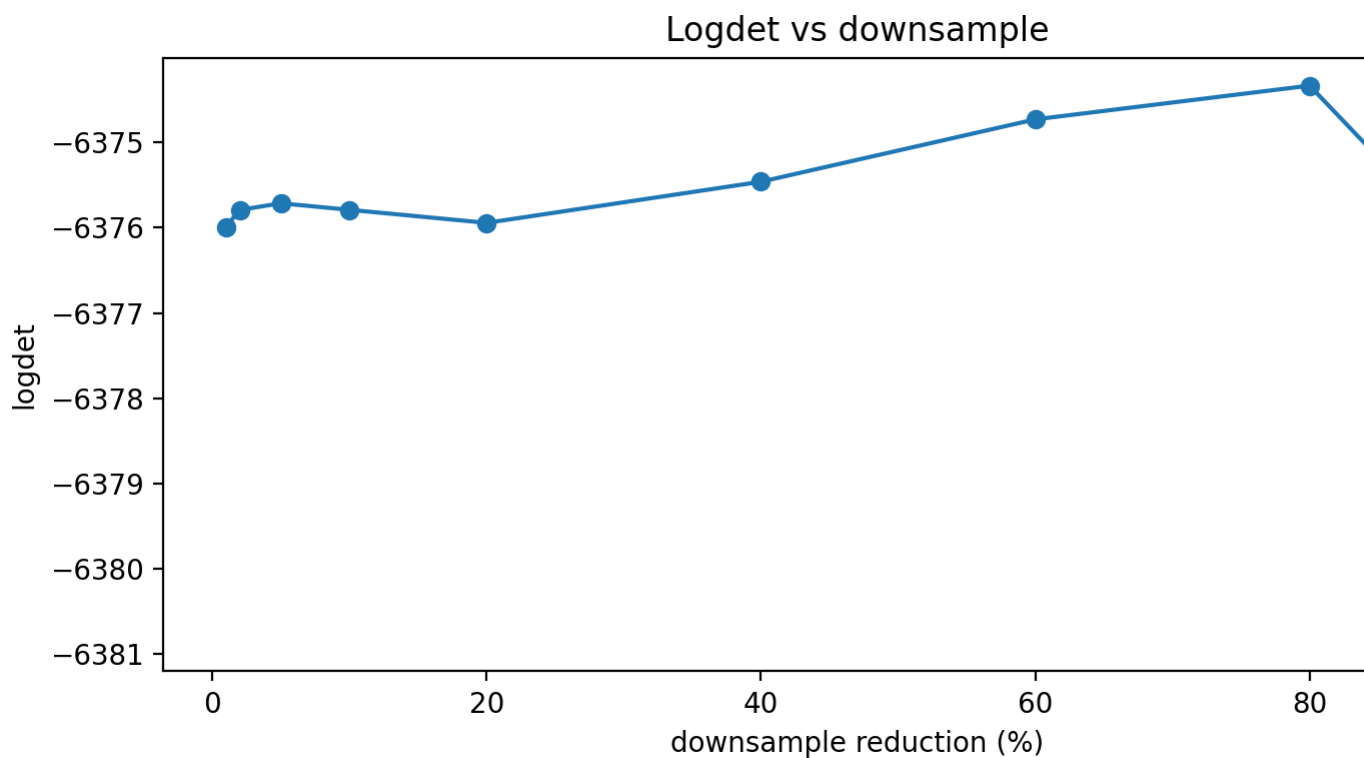
Downsampling beyond 60% sharply increases trace (e.g., 93% reduction $\hat{\Gamma}^n$ trace 12.83) while logdet falls to -6380.87, indicating uncertainty balloons along a handful of directions once spatial detail is largely removed.



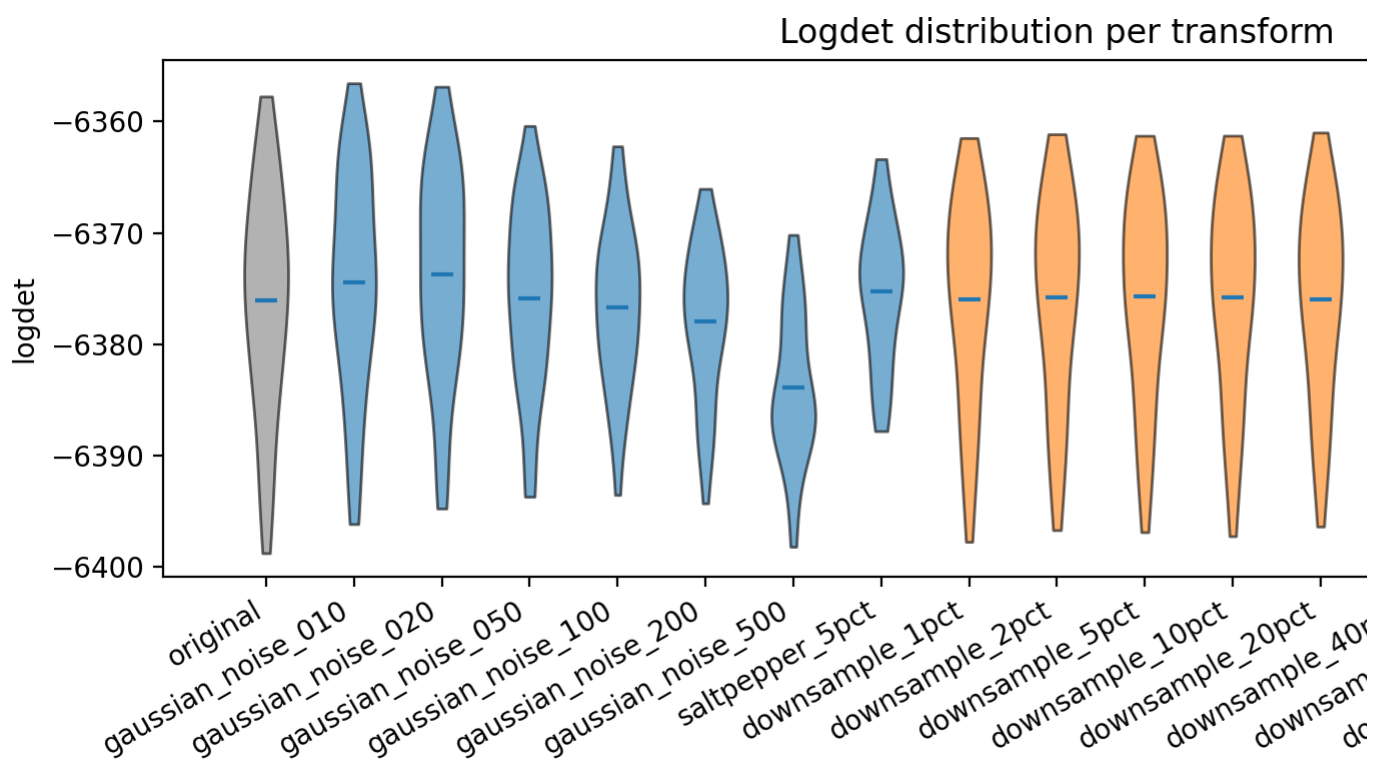
Trace distribution per transform



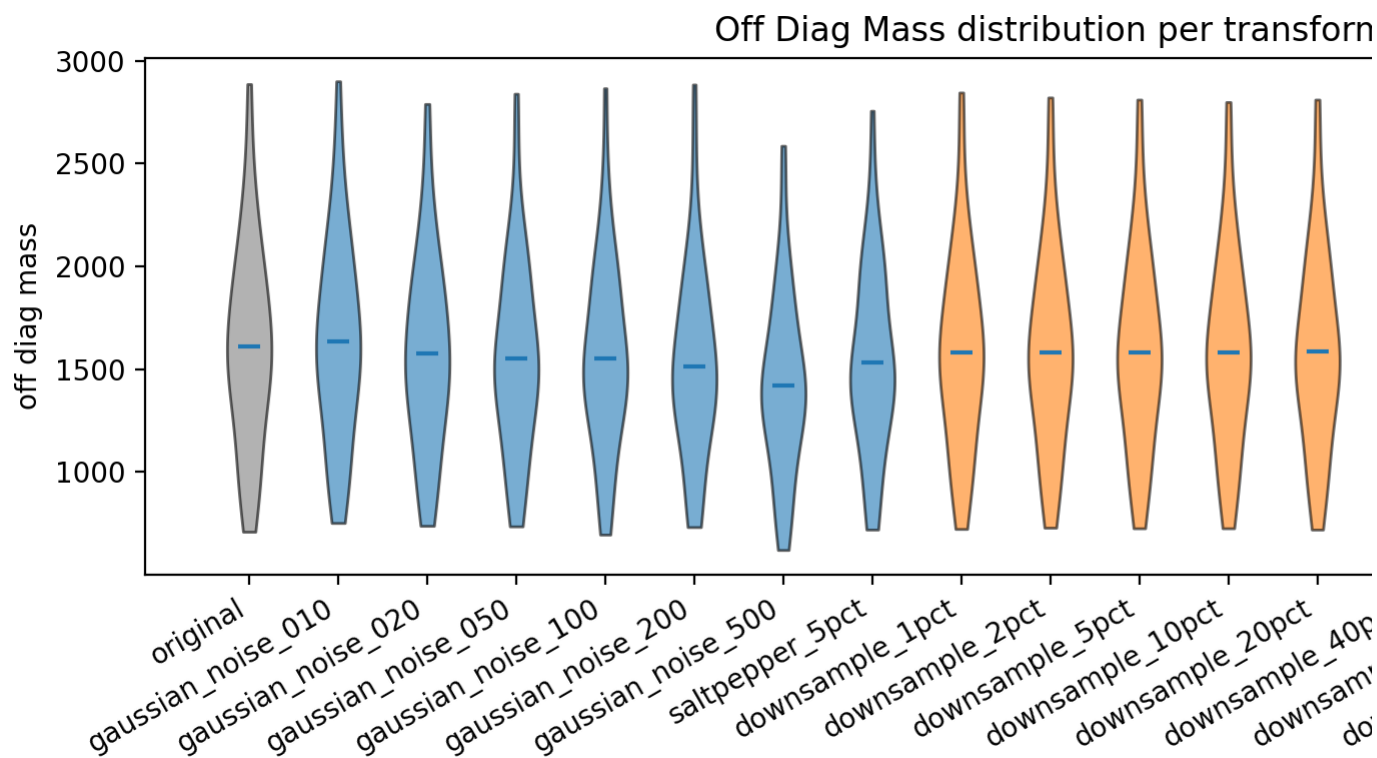
Trace vs downsampling severity



Logdet vs downsampling severity

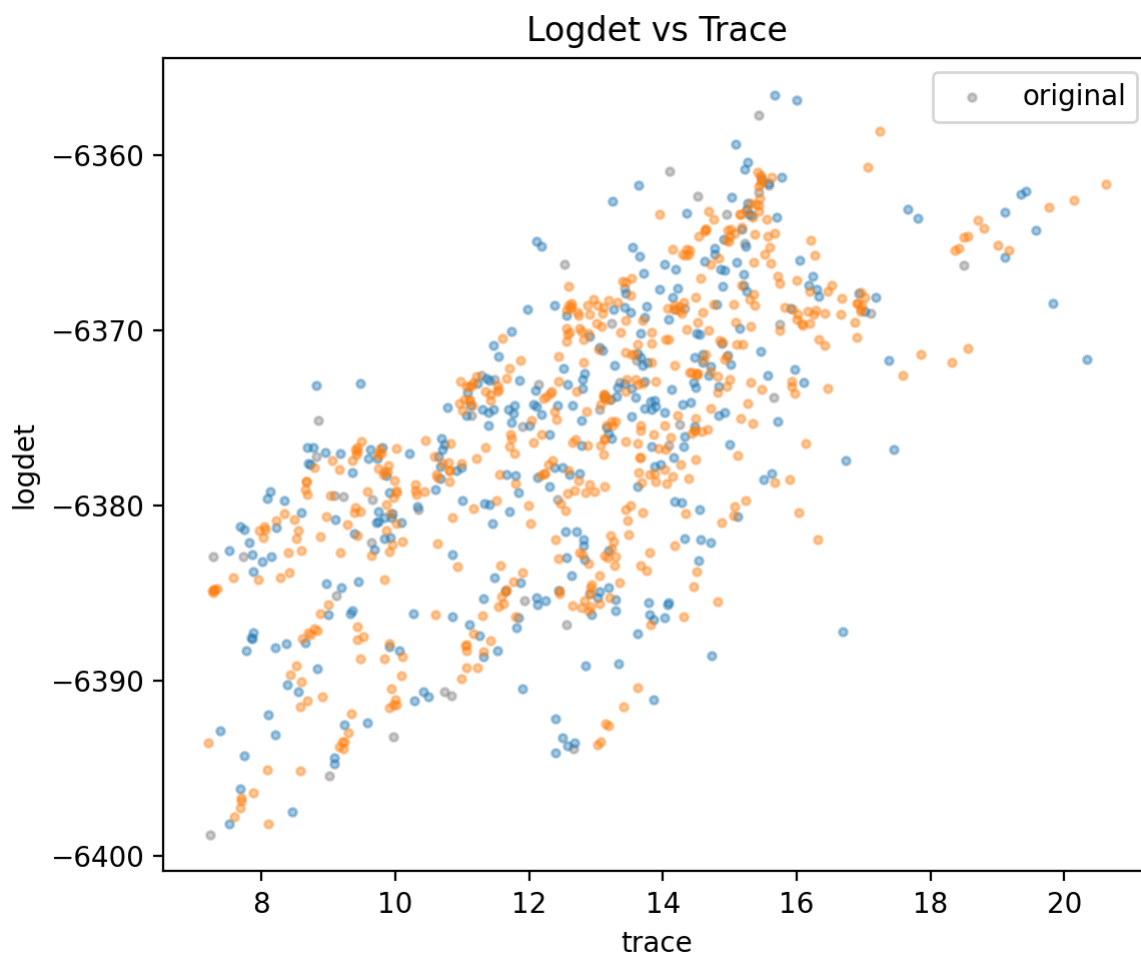


Logdet distribution per transform



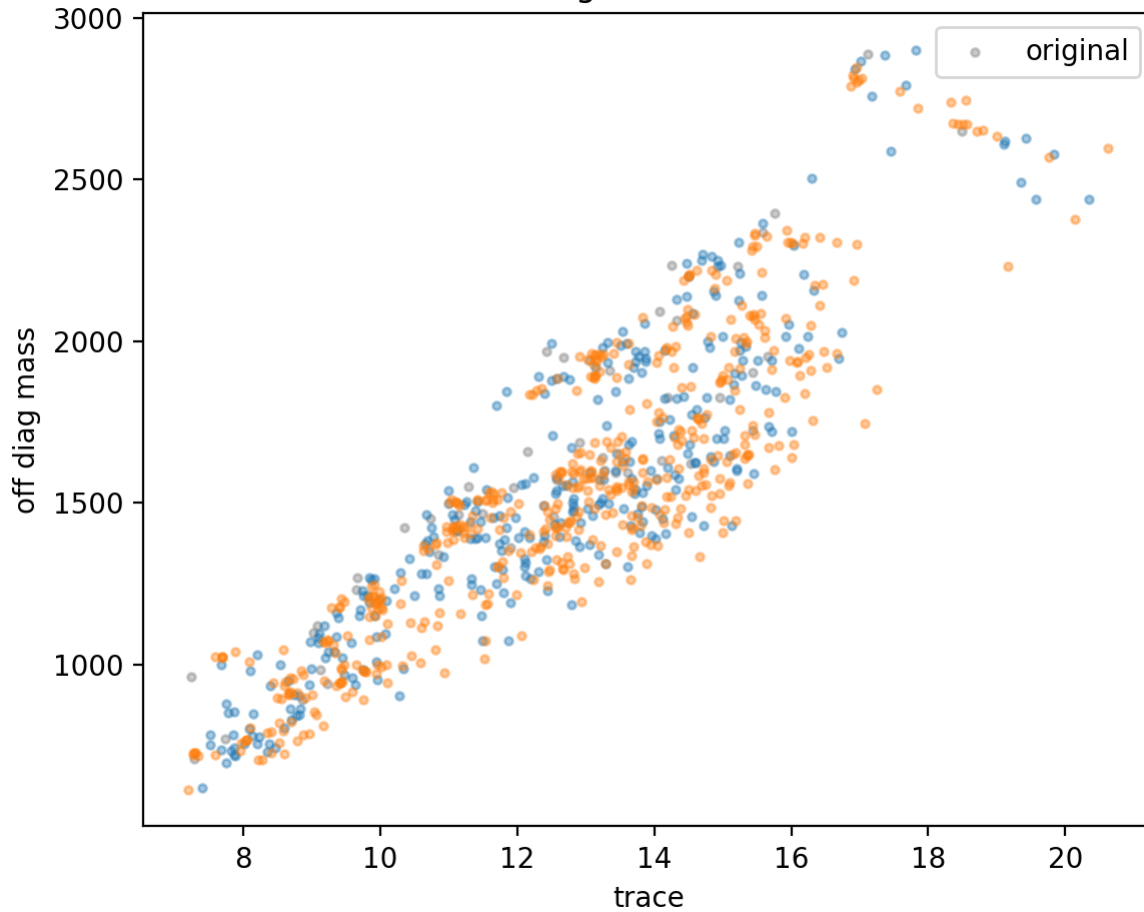
Off-diagonal mass distribution per transform

8. Cross-Metric Geometry



Logdet vs Trace scatter

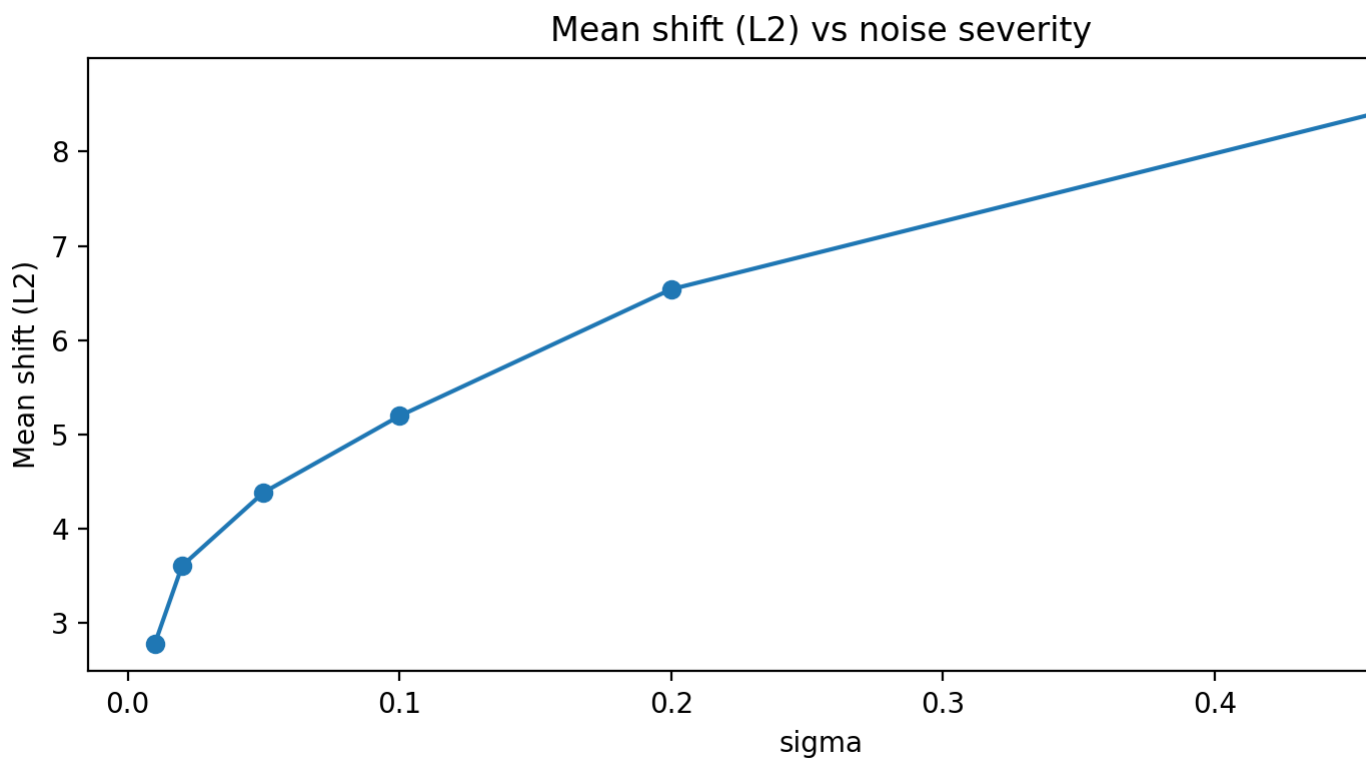
Off Diag Mass vs Trace



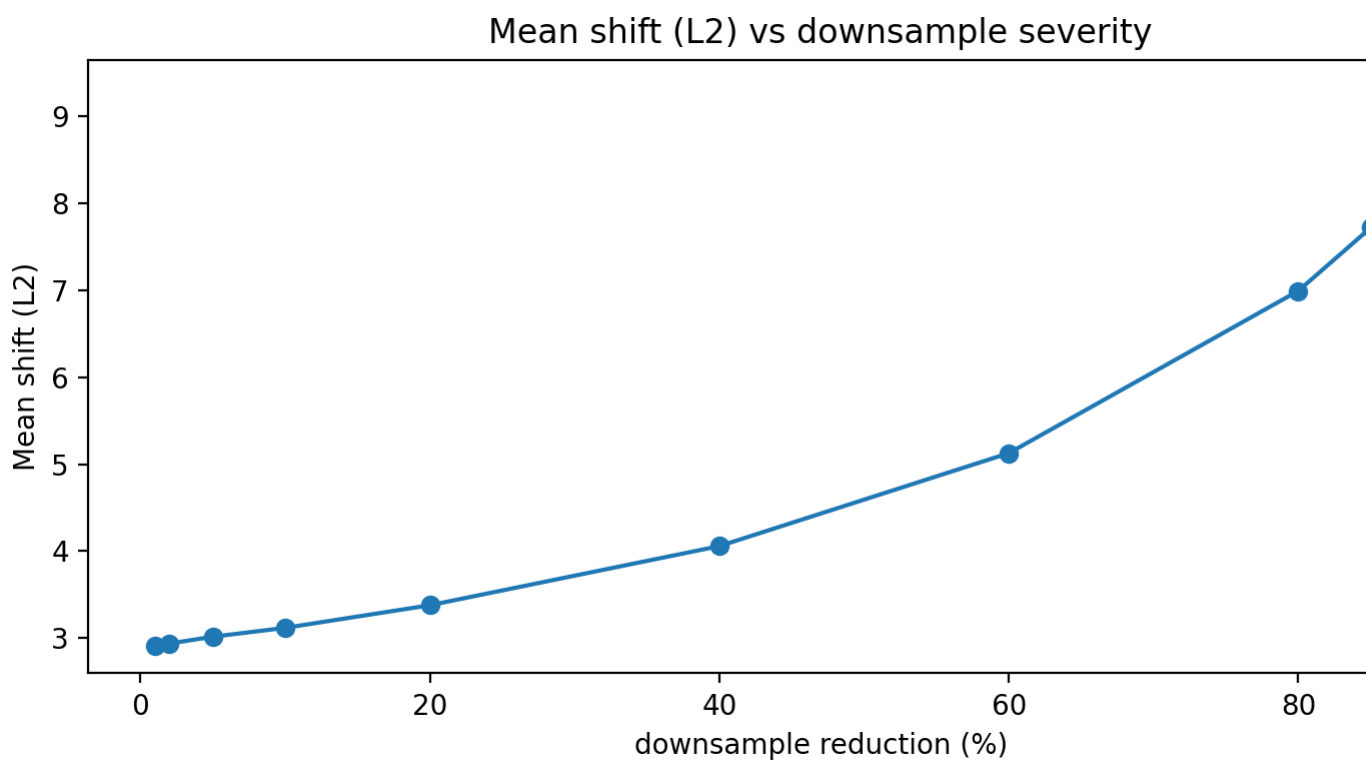
Off-diagonal mass vs Trace scatter

9. Mean Shift Diagnostics

transform	L2 shift	Mahalanobis shift
original	1.5934 Å± 0.2906	806.9347 Å± 97.6104
gaussian_noise_010	2.7813 Å± 0.6508	1666.2401 Å± 255.5203
gaussian_noise_020	3.6054 Å± 0.6817	2227.3307 Å± 358.3126
gaussian_noise_050	4.3841 Å± 0.6772	2788.7280 Å± 342.4279
gaussian_noise_100	5.1952 Å± 0.9378	3367.2794 Å± 419.5660
gaussian_noise_200	6.5398 Å± 1.0501	4228.4310 Å± 535.9685
gaussian_noise_500	8.7011 Å± 0.8827	5537.6085 Å± 582.0240
saltpepper_5pct	5.8839 Å± 0.9645	3828.5899 Å± 468.3757
downsample_1pct	2.9156 Å± 0.7442	1763.3735 Å± 465.4738
downsample_2pct	2.9370 Å± 0.7538	1767.1616 Å± 482.2413
downsample_5pct	3.0156 Å± 0.7507	1832.2940 Å± 465.8001
downsample_10pct	3.1176 Å± 0.7988	1908.4585 Å± 494.0486
downsample_20pct	3.3776 Å± 0.8529	2091.0686 Å± 520.2487
downsample_40pct	4.0572 Å± 0.8832	2561.1259 Å± 518.2966
downsample_60pct	5.1260 Å± 0.8931	3215.4408 Å± 481.4732
downsample_80pct	6.9914 Å± 0.7538	4419.8093 Å± 534.7429
downsample_85pct	7.7168 Å± 0.6615	4915.8207 Å± 549.4709
downsample_90pct	8.8262 Å± 0.6330	5610.9697 Å± 584.7671
downsample_93pct	9.3279 Å± 0.6722	5948.2206 Å± 621.6978

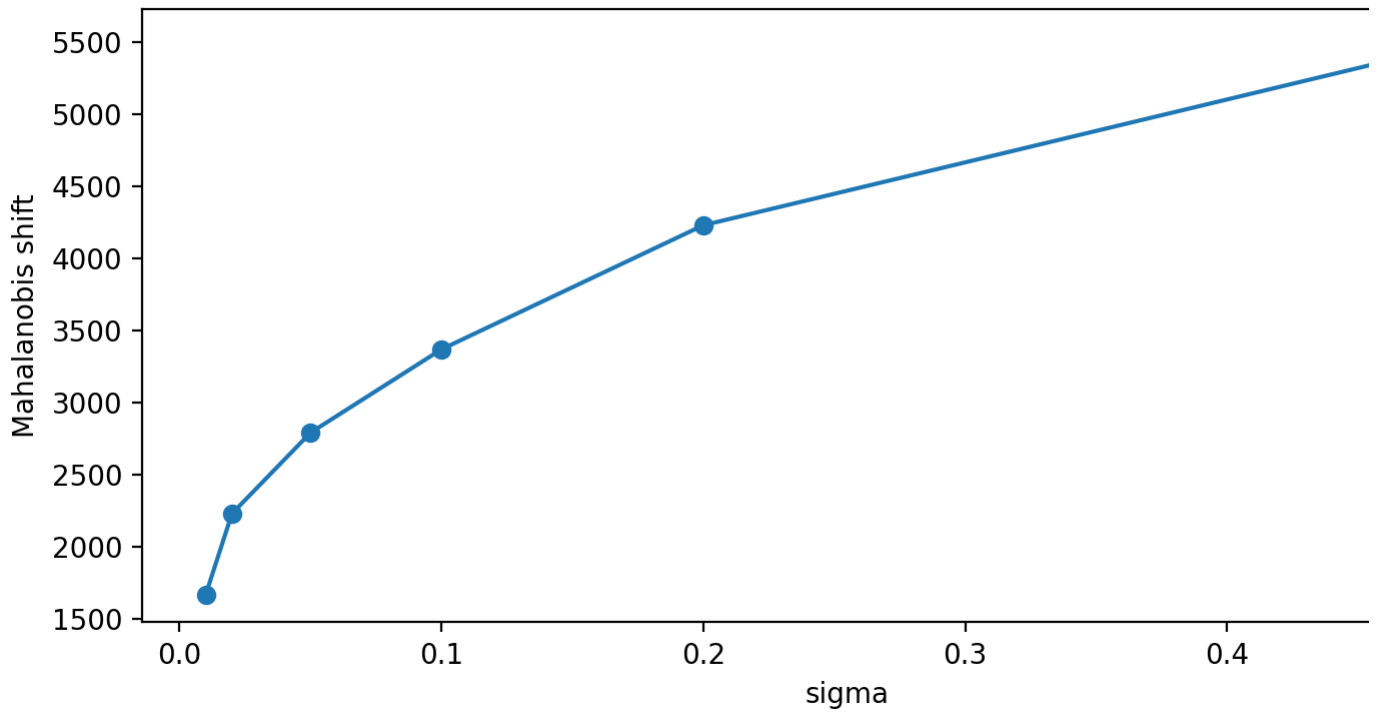


Mean shift vs noise severity



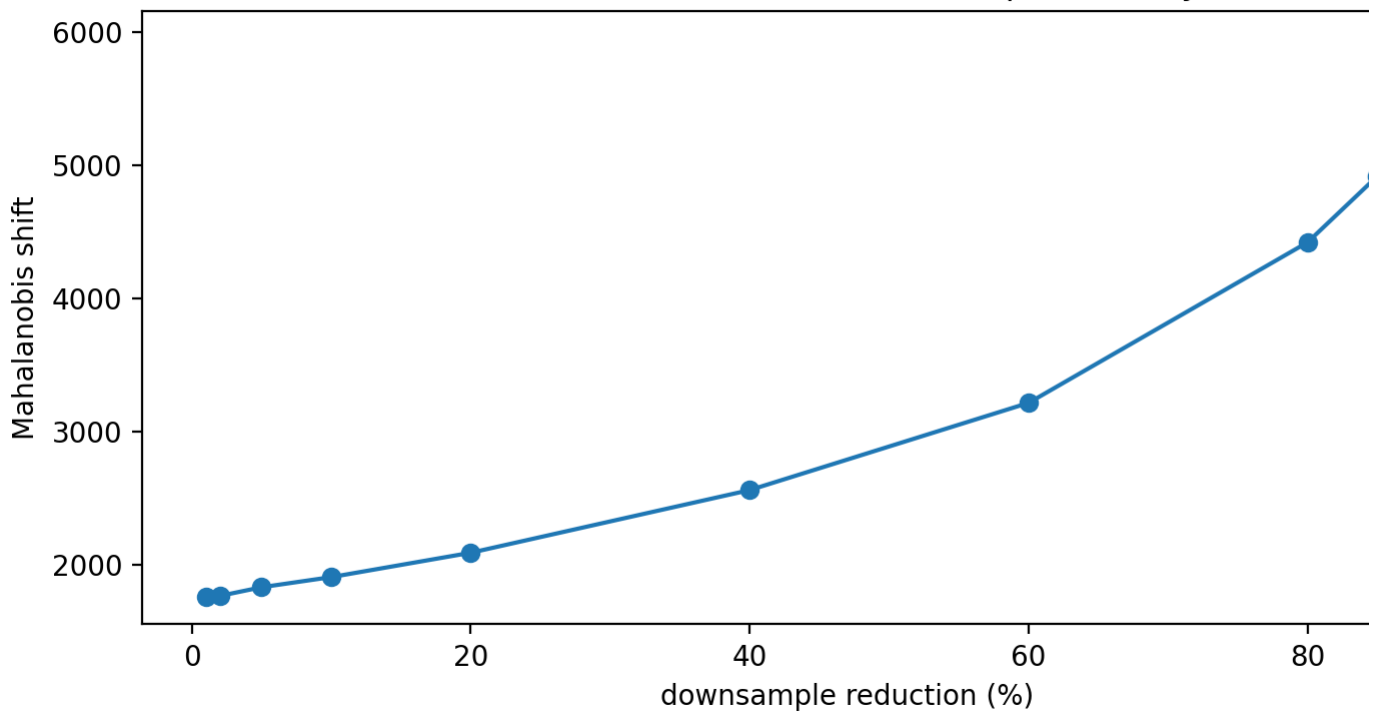
Mean shift vs downsampling severity

Mahalanobis shift vs noise severity



Mahalanobis shift vs noise severity

Mahalanobis shift vs downsample severity

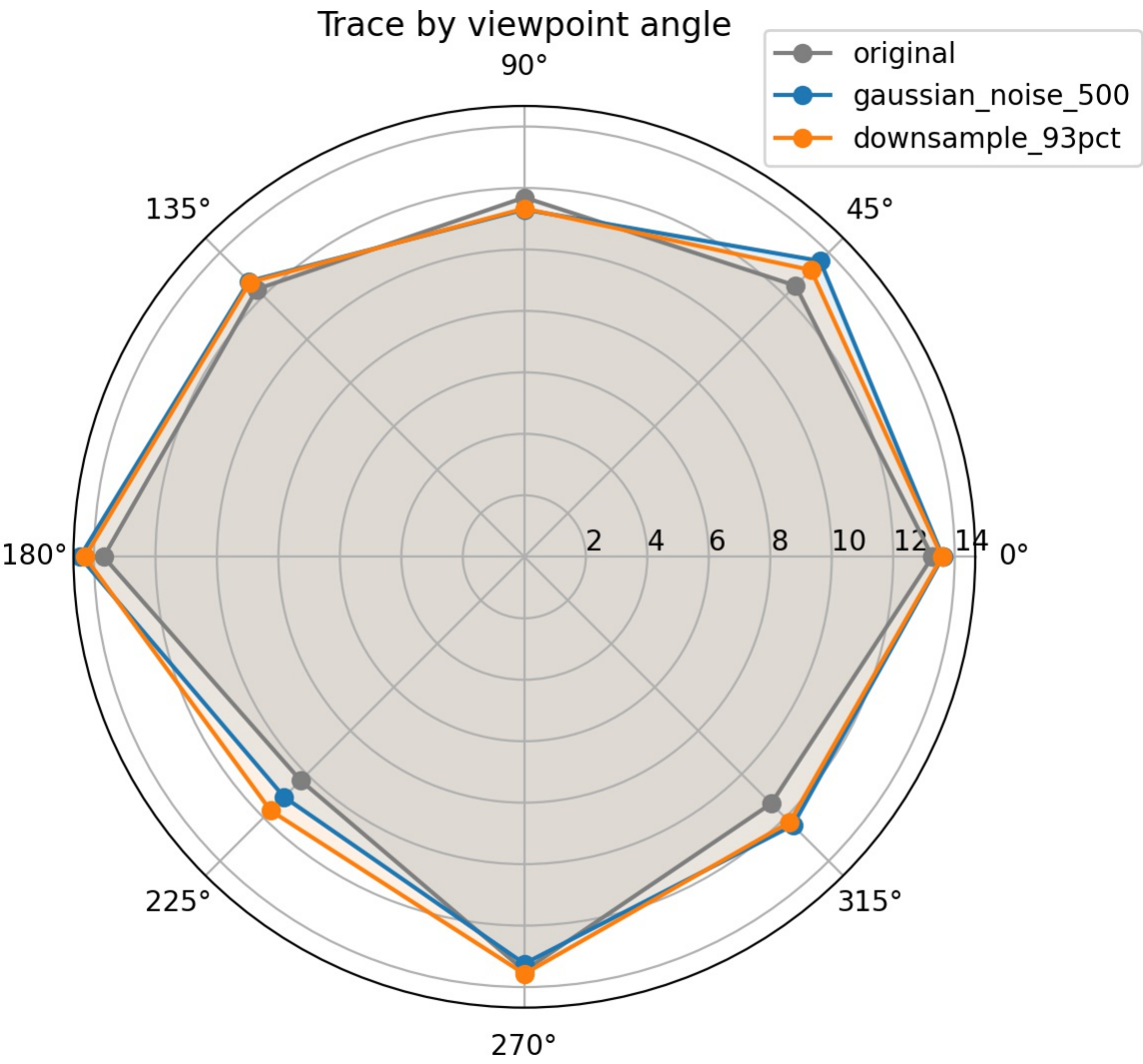


Mahalanobis shift vs downsampling severity

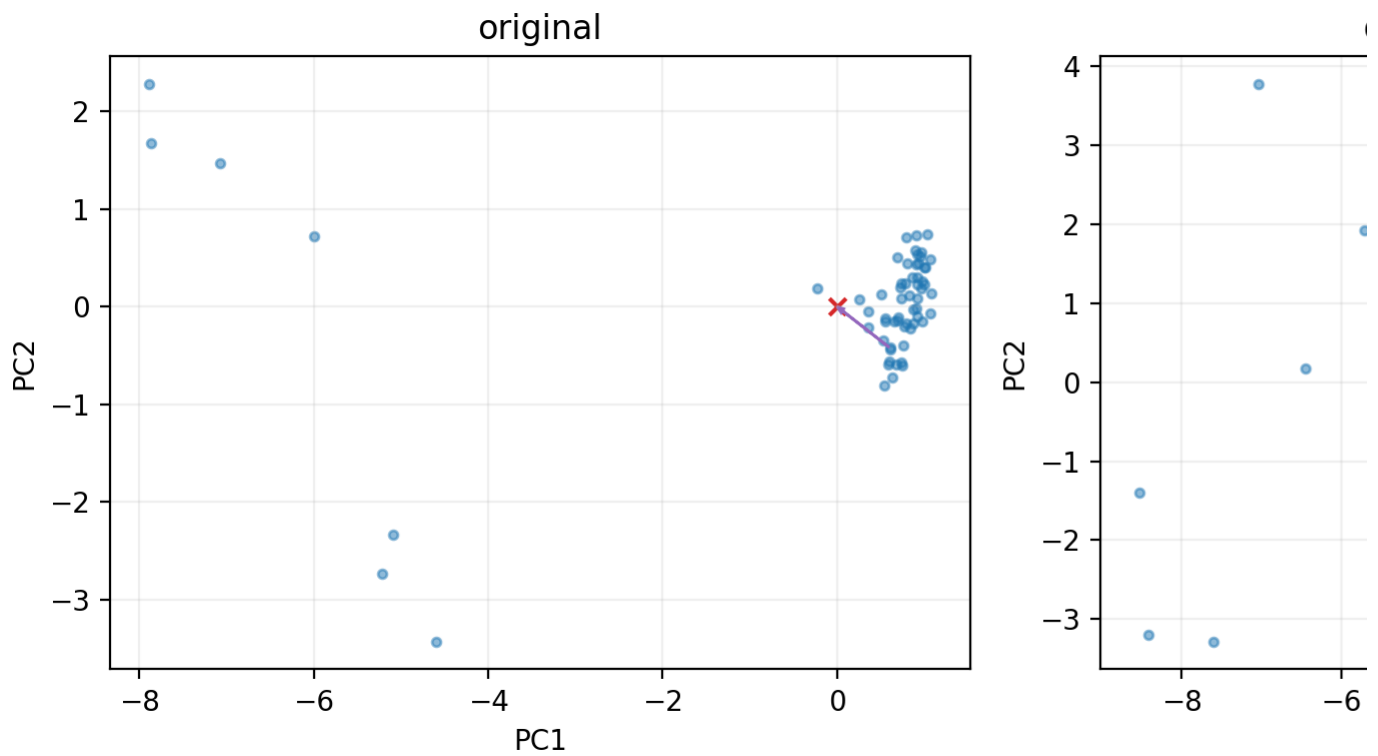
10. Spectral & Tangent Geometry

transform	tangent trace	tangent $\hat{f}_{\max}$	spectral entropy	top-10 share	circular variance
original	8.5239 $\hat{A} \pm 1.6899$	2.2936 $\hat{A} \pm 0.9171$	2.5811 $\hat{A} \pm 0.2383$	0.7565 $\hat{A} \pm 0.0436$	0.0633 $\hat{A} \pm 0.0157$
gaussian_noise_010	8.7489 $\hat{A} \pm 1.7260$	2.3569 $\hat{A} \pm 0.9274$	2.5859 $\hat{A} \pm 0.2303$	0.7557 $\hat{A} \pm 0.0424$	0.0654 $\hat{A} \pm 0.0161$
gaussian_noise_020	8.7992 $\hat{A} \pm 1.6985$	2.3006 $\hat{A} \pm 0.9398$	2.6201 $\hat{A} \pm 0.2280$	0.7518 $\hat{A} \pm 0.0420$	0.0631 $\hat{A} \pm 0.0155$
gaussian_noise_050	8.5107 $\hat{A} \pm 1.6004$	2.2645 $\hat{A} \pm 0.9120$	2.5796 $\hat{A} \pm 0.2205$	0.7574 $\hat{A} \pm 0.0400$	0.0617 $\hat{A} \pm 0.0152$
gaussian_noise_100	8.4447 $\hat{A} \pm 1.5670$	2.2813 $\hat{A} \pm 0.9015$	2.5460 $\hat{A} \pm 0.2187$	0.7620 $\hat{A} \pm 0.0389$	0.0625 $\hat{A} \pm 0.0152$
gaussian_noise_200	8.2401 $\hat{A} \pm 1.3829$	2.1899 $\hat{A} \pm 0.8128$	2.5005 $\hat{A} \pm 0.2321$	0.7685 $\hat{A} \pm 0.0389$	0.0601 $\hat{A} \pm 0.0144$
gaussian_noise_500	7.7364 $\hat{A} \pm 1.4081$	2.0397 $\hat{A} \pm 0.7857$	2.3501 $\hat{A} \pm 0.2266$	0.7948 $\hat{A} \pm 0.0384$	0.0552 $\hat{A} \pm 0.0146$
saltpopper_5pct	8.6093 $\hat{A} \pm 1.3903$	2.2036 $\hat{A} \pm 0.8332$	2.5686 $\hat{A} \pm 0.2253$	0.7608 $\hat{A} \pm 0.0370$	0.0626 $\hat{A} \pm 0.0142$
downsample_1pct	8.6512 $\hat{A} \pm 1.6931$	2.3227 $\hat{A} \pm 0.9141$	2.5873 $\hat{A} \pm 0.2266$	0.7572 $\hat{A} \pm 0.0406$	0.0619 $\hat{A} \pm 0.0148$
downsample_2pct	8.6612 $\hat{A} \pm 1.6923$	2.3173 $\hat{A} \pm 0.9090$	2.5889 $\hat{A} \pm 0.2245$	0.7570 $\hat{A} \pm 0.0402$	0.0620 $\hat{A} \pm 0.0149$
downsample_5pct	8.6939 $\hat{A} \pm 1.6944$	2.3229 $\hat{A} \pm 0.9144$	2.5880 $\hat{A} \pm 0.2230$	0.7574 $\hat{A} \pm 0.0400$	0.0620 $\hat{A} \pm 0.0147$
downsample_10pct	8.6884 $\hat{A} \pm 1.7005$	2.3288 $\hat{A} \pm 0.9098$	2.5866 $\hat{A} \pm 0.2215$	0.7574 $\hat{A} \pm 0.0397$	0.0620 $\hat{A} \pm 0.0147$
downsample_20pct	8.7053 $\hat{A} \pm 1.6775$	2.3396 $\hat{A} \pm 0.9162$	2.5793 $\hat{A} \pm 0.2160$	0.7591 $\hat{A} \pm 0.0387$	0.0620 $\hat{A} \pm 0.0147$
downsample_40pct	8.8511 $\hat{A} \pm 1.6547$	2.3567 $\hat{A} \pm 0.9101$	2.5755 $\hat{A} \pm 0.2080$	0.7617 $\hat{A} \pm 0.0360$	0.0625 $\hat{A} \pm 0.0144$
downsample_60pct	8.9885 $\hat{A} \pm 1.5274$	2.3026 $\hat{A} \pm 0.8298$	2.5678 $\hat{A} \pm 0.1983$	0.7650 $\hat{A} \pm 0.0341$	0.0626 $\hat{A} \pm 0.0141$

downsample_80pct	9.0825 Å ± 1.4520	2.2022 Å ± 0.7401	2.5403 Å ± 0.2085	0.7721 Å ± 0.0280	0.0598 Å ± 0.0141
downsample_85pct	8.8670 Å ± 1.4370	2.1181 Å ± 0.7449	2.5217 Å ± 0.2287	0.7726 Å ± 0.0357	0.0585 Å ± 0.0141
downsample_90pct	8.6496 Å ± 1.3900	2.1075 Å ± 0.7272	2.5033 Å ± 0.2229	0.7787 Å ± 0.0332	0.0573 Å ± 0.0138
downsample_93pct	8.2983 Å ± 1.4501	2.0909 Å ± 0.7254	2.4687 Å ± 0.2265	0.7821 Å ± 0.0337	0.0577 Å ± 0.0140



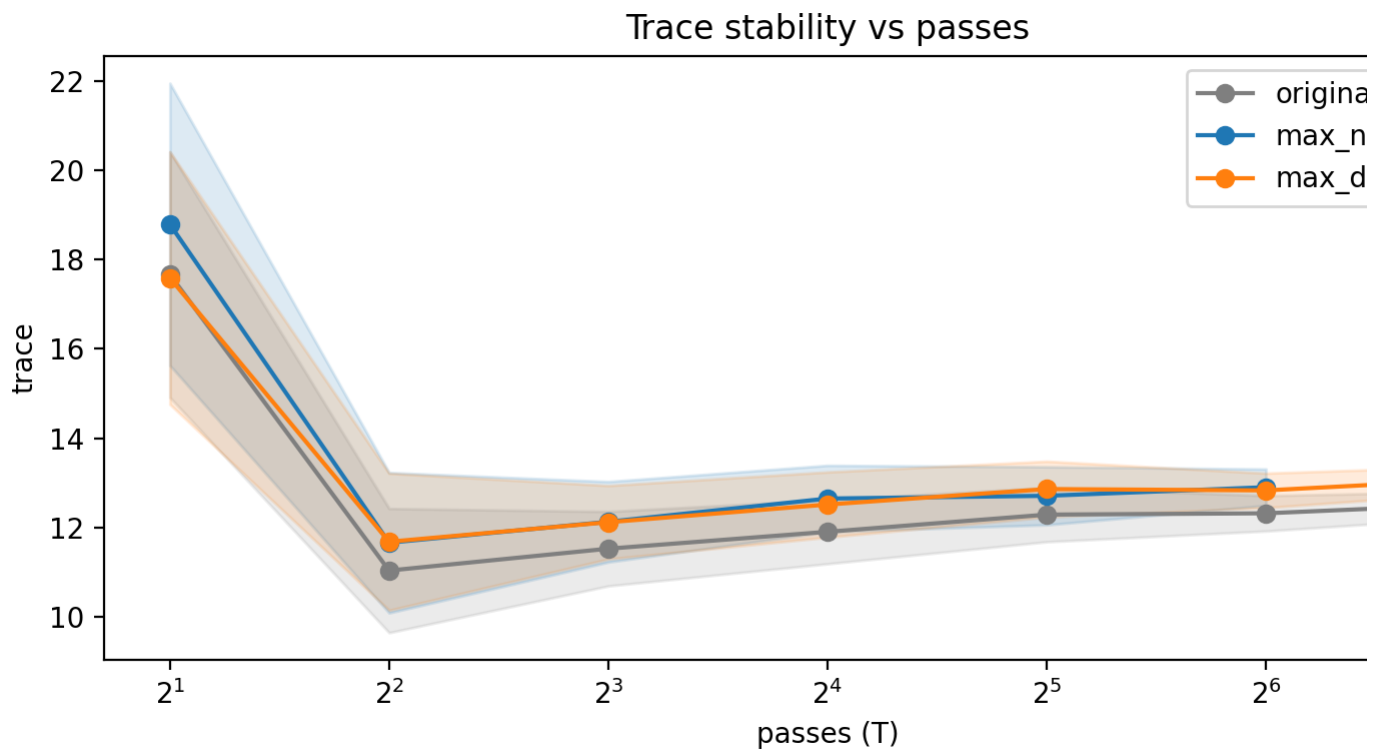
Trace by viewpoint angle



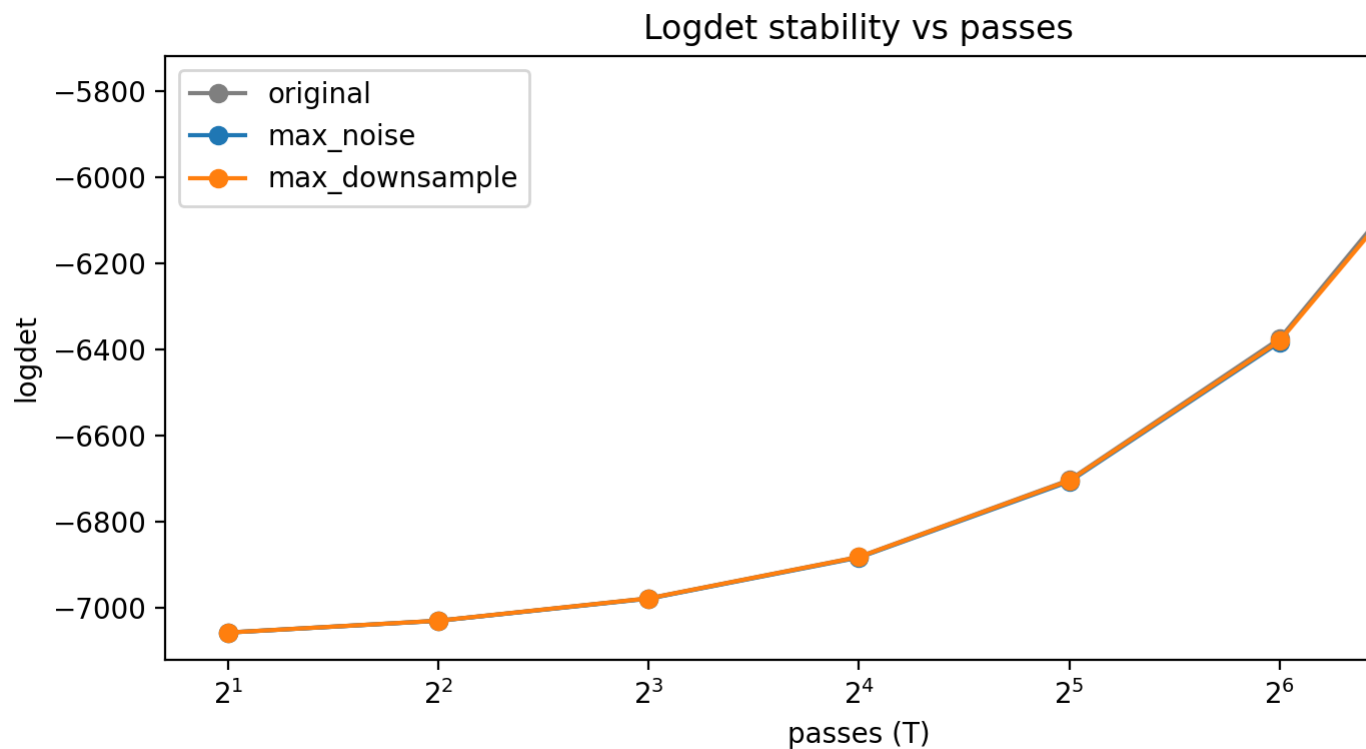
PCA gallery for representative sample

11. Pass Count Stability

We evaluate trace and log-determinant stability across Monte Carlo pass counts $T \in \{2, 4, 8, 16, 32, 64, 128\}$. Lower T increases estimator noise; curves should flatten as T grows. See stability plots in this section if generated.



Trace stability vs passes



Logdet stability vs passes

12. Discussion & Outlook

- **High noise ($\hat{f}=0.50$)** expands trace by +4.0% and drops logdet by 7.89. The stochastic mean drifts 8.70 L2 units from the deterministic embedding, and tangent variance settles around 7.74, confirming broader but directional uncertainty.
- **Extreme downsampling (93% $\hat{t}$ 16px)** shifts trace by +4.2% and drops logdet by 4.84. Mahalanobis drift reaches 5948.2, while spectral entropy averages 2.47, signalling variance concentration into fewer modes.
- **Viewpoint sensitivity:** At 225°, trace shifts by +1.38 relative to the clean view, indicating that certain orientations (e.g. flank perspectives) become the most uncertain once detail is removed.
- **Pass-count stability:** Trace estimates converge within $\hat{A}\pm 3.319$ by 32 passes; high-noise and heavy-downsample settings widen the band only modestly, so $T=64$ remains a safe budget.
- **Predictive head:** Mutual information stays near $2.54e-05$ for all conditions, so predictive entropy contributes little insight compared with embedding-space diagnostics.

13. Class-Level Trace Shifts

	class	trace (original)	trace (saltpepper_5pct)	$\hat{t}$ noise	trace (downsample_93pct)	$\hat{t}$ downsample
	Indigo	11.3669 $\hat{A}\pm 1.6120$	11.7008 $\hat{A}\pm 1.3592$	0.3339	12.2583 $\hat{A}\pm 1.2720$	0.8914
	Magenta	11.9280 $\hat{A}\pm 3.2723$	12.0372 $\hat{A}\pm 3.4081$	0.1091	12.6884 $\hat{A}\pm 3.7444$	0.7603
	Moose3	12.6069 $\hat{A}\pm 3.3034$	14.2672 $\hat{A}\pm 3.0083$	1.6603	13.4894 $\hat{A}\pm 3.3945$	0.8825
	White	13.1045 $\hat{A}\pm 2.9819$	12.9342 $\hat{A}\pm 2.4708$	-0.1703	13.2767 $\hat{A}\pm 2.5596$	0.1722
	Wolf2	11.7160 $\hat{A}\pm 2.8515$	12.8059 $\hat{A}\pm 2.5326$	1.0899	12.2501 $\hat{A}\pm 2.4074$	0.5341
	Yellow	13.2319 $\hat{A}\pm 2.3014$	13.0068 $\hat{A}\pm 2.1829$	-0.2251	13.1653 $\hat{A}\pm 2.5896$	-0.0666

14. Example Perturbations



- Unmodified image:



- Additive Gaussian noise ($\sigma = 0.01$):



- Additive Gaussian noise ($\sigma = 0.02$):



- Additive Gaussian noise ($\sigma = 0.05$):



- Additive Gaussian noise ($\sigma = 0.1$):



- Additive Gaussian noise ($\sigma = 0.2$):



- Additive Gaussian noise ($\sigma = 0.5$):



- Salt & pepper noise (5% pixels, 60% salt):



- Downsample to 222px (encoder base 224px, 1% reduction) then upscale:



- Downsample to 220px (encoder base 224px, 2% reduction) then upscale:



- Downsample to 213px (encoder base 224px, 5% reduction) then upscale:



- Downsample to 202px (encoder base 224px, 10% reduction) then upscale:



- Downsample to 179px (encoder base 224px, 20% reduction) then upscale:



- Downsample to 134px (encoder base 224px, 40% reduction) then upscale:



- Downsample to 90px (encoder base 224px, 60% reduction) then upscale:



- Downsample to 45px (encoder base 224px, 80% reduction) then upscale:



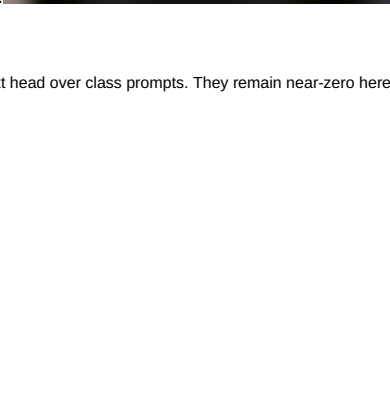
- Downsample to 34px (encoder base 224px, 85% reduction) then upscale:



- Downsample to 22px (encoder base 224px, 90% reduction) then upscale:

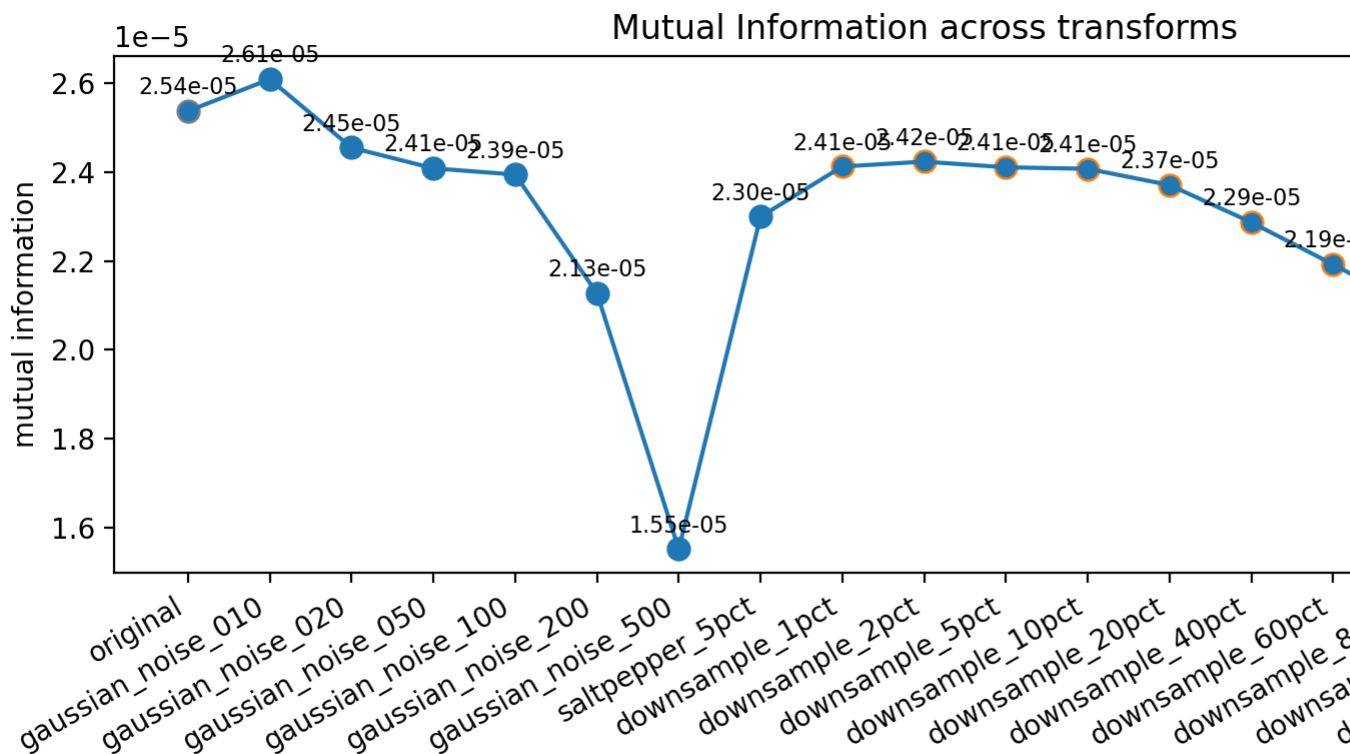


- Downsample to 16px (encoder base 224px, 93% reduction) then upscale:

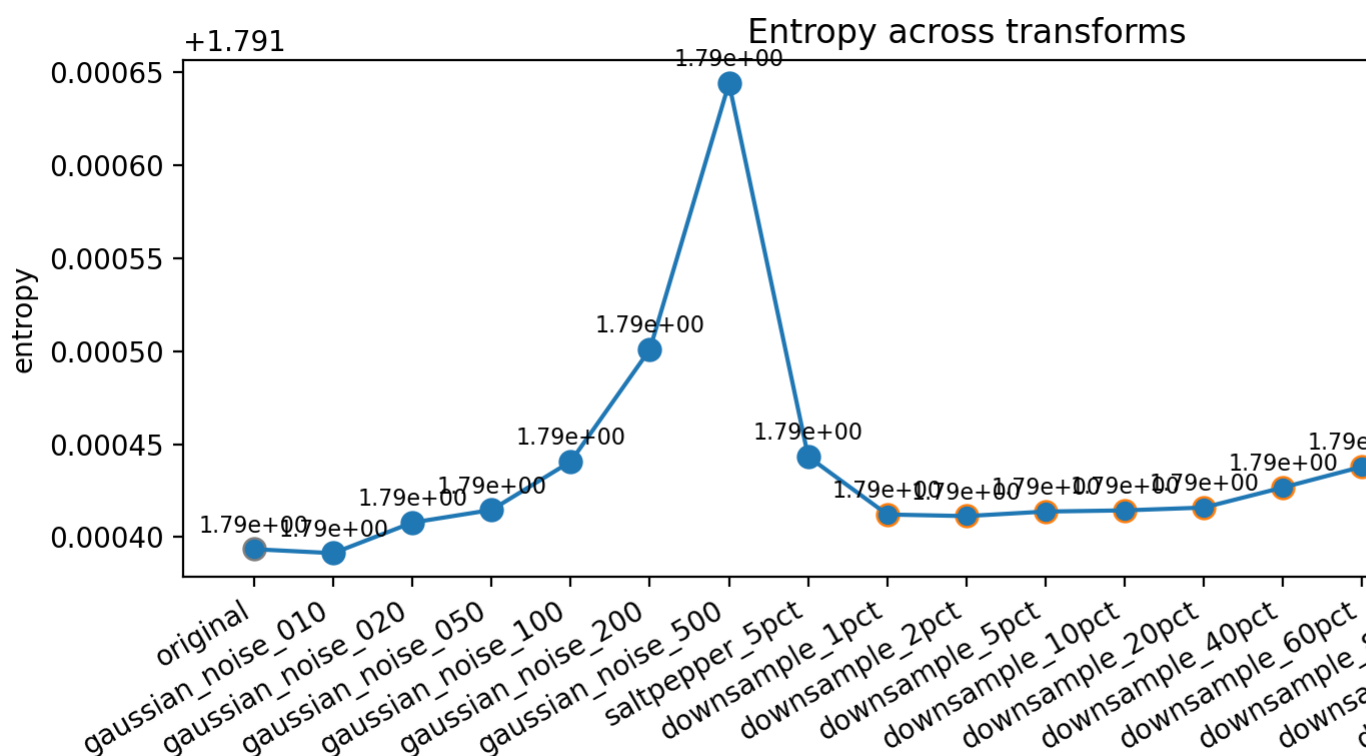


Appendix: Predictive Diagnostics (MI & Entropy)

Predictive mutual information (epistemic) and entropy are derived from the CLIP text head over class prompts. They remain near-zero here due to prompt unanimity and low dropout; we include them for completeness.



Mutual information across transforms



Entropy across transforms